

## Naive Bayes Classifier

Introduction: It is a probabilistic supervised ML algorithm mainly used for classification tasks.

Intuition: Think of it as making a 'yes/no' decision based on past data, like spam detection.

Bayes' Theorem:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- $P(A|B)$ : Posterior (Probability of A given B)
- $P(B|A)$ : Likelihood (Probability of B given A)
- $P(A)$ : Prior (Initial probability of A)
- $P(B)$ : Evidence (Total probability of B)

Conditional Independence Assumption:

Naive Bayes assumes features are conditionally independent given the class label.

$$P(A \cap B) = P(A) \cdot P(B)$$

↳ Why? This assumption significantly simplifies the computation, making the algorithm fast.

Mapping to Machine Learning:

$$P(\text{output} / \text{input}) \rightarrow P(y / x)$$

Here, 'y' is the class label (output), and 'x' represents the features (input).

Multiple Feature Case:

Data: Features  $x_1, x_2, x_3$  and class y

$$P(y | x_1, x_2, x_3) = \frac{P(x_1, x_2, x_3 | y) * P(y)}{P(x_1, x_2, x_3)} \begin{matrix} \rightarrow A \\ \rightarrow B \end{matrix}$$



## Naive Bayes - Numerical Example

### Given Dataset:

| Weather ( $x_1$ ) | Wind ( $x_2$ ) | Play ( $y$ ) |
|-------------------|----------------|--------------|
| Hot               | Weak           | Yes          |
| Hot               | Strong         | No           |
| Cold              | Weak           | Yes          |
| Cold              | Strong         | No           |
| Hot               | Weak           | Yes          |

(Note: Sample data to match subsequent calculations)

### Step 1: Prior Probabilities

Total samples = 5

$P(\text{Yes}) = \frac{3}{5} = 0.6$  (Prior probability of Play=Yes)

$P(\text{No}) = \frac{2}{5} = 0.4$  (Prior probability of Play=No)

### Step 2: Conditional Probabilities

For Weather ( $x_1$ ):

$$P(\text{Hot} | \text{Yes}) = \frac{2}{3}$$

$$P(\text{Cold} | \text{Yes}) = \frac{1}{3}$$

$$P(\text{Hot} | \text{No}) = \frac{1}{2}$$

$$P(\text{Cold} | \text{No}) = \frac{1}{2}$$

For Wind ( $x_2$ ):

$$P(\text{Weak} | \text{Yes}) = \frac{2}{3}$$

$$P(\text{Strong} | \text{Yes}) = \frac{1}{3}$$

$$P(\text{Weak} | \text{No}) = \frac{0}{2} = 0$$

$$P(\text{Strong} | \text{No}) = \frac{2}{2} = 1$$

### Step 3: Apply Naive Bayes Formula

Query: Weather=Hot, Wind=Strong. Will they play?

$$\begin{aligned}
 P(\text{Yes} | \text{Hot}, \text{Strong}) &\propto P(\text{Hot} | \text{Yes}) \times P(\text{Strong} | \text{Yes}) \times P(\text{Yes}) \\
 &= \left(\frac{2}{3}\right) \times \left(\frac{1}{3}\right) \times \left(\frac{3}{5}\right)
 \end{aligned}$$

$$\begin{aligned}
 P(\text{No} | \text{Hot}, \text{Strong}) &\propto P(\text{Hot} | \text{No}) \times P(\text{Strong} | \text{No}) \times P(\text{No}) \\
 &= \left(\frac{1}{2}\right) \times (1) \times \left(\frac{2}{5}\right)
 \end{aligned}$$

### Step 4: Final Calculation & Comparison

$$P(\text{Yes} | \text{Hot}, \text{Strong}) \propto \frac{6}{45} \approx 0.133$$

$$P(\text{No} | \text{Hot}, \text{Strong}) \propto \frac{2}{10} = 0.2$$

Since  $0.2 > 0.133$ ,  $P(\text{No} | \text{Hot}, \text{Strong}) > P(\text{Yes} | \text{Hot}, \text{Strong})$ .

### Final Conclusion:

Since  $P(\text{No} | \text{Hot}, \text{Strong})$  is higher, the predicted class is NO.



## Types of Naive Bayes Classifier

Different types of Naive Bayes are used based on the type of data. The choice depends on whether data is continuous, count-based, or binary.

### 1. Gaussian Naive Bayes

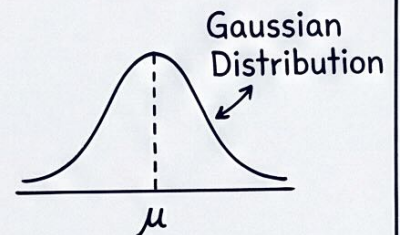
Used when features are continuous (real-valued).

Assumes data follows a normal (Gaussian) distribution.

$$P(x|y) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Here,  $\mu$  = mean,  $\sigma$  = standard deviation.

e.g., Height, weight, temperature, marks.



### 2. Multinomial Naive Bayes

Used when features represent counts or frequencies.

Commonly used in text classification.

e.g., Number of times a word appears in a document (spam detection).



Word Counts in Document

Works well when features are non-negative integers.

### 3. Bernoulli Naive Bayes

Used when features are binary (0 or 1).

Only checks presence or absence of a feature.

e.g., Word present or not present in email.

Yes / No type data.

| Feature Value | Meaning       |
|---------------|---------------|
| 0 →           | No (Absent)   |
| 1 →           | Yes (Present) |

|                                                                           |                                |
|---------------------------------------------------------------------------|--------------------------------|
| <u>Comparison</u> : Gaussian → Continuous data<br>Bernoulli → Binary data | Multinomial → Count-based data |
|---------------------------------------------------------------------------|--------------------------------|